Muhammad Faras Siddiqui

+971 56 881 5205 / +92 335 904 8481
linkedin.com/in/muhammad-faras-siddiqui
siddiquifaras@gmail.com
github.com/siddiquifaras
gitlab.com/sidfaras
Islamabad, Pakistan

Summary

I build and deploy applied AI systems that perform in both conventional and non-conventional environments, from cloud GPUs to air-gapped networks and edge hardware. My work focuses on AI infrastructure: agent architectures, LLM inference and serving, evaluation, and the deployment pipelines that keep it all reliable in production.

Experience

- **Founding Engineer & Technical Lead** — Feb 2025 – Present
  CVision — *Wyoming, USA (Remote)*
  - **Revenue & Client Impact**: Built CVision's first distributed AI systems for industrial and SME clients, bringing in **$10K+** in month one and **$150K+** in year one, and contributing to **$2M+** in deal value.
  - **Constrained Deployments**: Lead design and deployment of production AI on edge, industrial and low-latency stacks, working with hardware teams under tight memory, compute and energy budgets.
  - **Agentic Systems**: Built and maintain the infrastructure behind CVision's agentic systems: retrieval-augmented LLMs, GNN-based reasoning and RevOps reporting over multiple sources of truth.
  - **Infrastructure & MLOps**: Set up core infrastructure from scratch: self-hosted runtimes, CI/CD, model packaging and real-time deployment, with fast rollout and recovery across live client deployments.
  - **Neuromorphic Tooling**: Delivered neuromorphic tooling as internal Python libraries, including compression that shrinks 10-layer networks from **200 KB to 200 B**.

- **AI Systems Engineer (Contract)** — Feb 2026 – Oct 2026
  Noah Labs — *San Francisco, USA (Remote)*
  - **Agent runtime**: Built core parts of the agent runtime for an air-gapped AI coding IDE (planning, parallel sub-agents, tool calling, memory, code retrieval) and fixed cross-agent blocking with workspace isolation.
  - **GPU inference**: Ran Gemma 4 31B on vLLM across **NVIDIA GH200** and **2× A100** GPUs, wired to the local IDE for end-to-end tests with real tool calls.
  - **Evaluation**: Built the agent evaluation harness: multi-seed golden-task runs on live GPU inference, scoring task completion, tool use, code quality and policy compliance to catch regressions before release.
  - **Security**: Integrated and evaluated data-loss prevention for autonomous agent workflows, and designed guardrails so untrusted content (documents, tickets, commits) can never trigger an agent action on its own.
  - **CI/CD**: Hardened CI/CD for the agent platform with production Docker images, integration smoke tests and parallelized test pipelines.

- **Deep Learning Research Intern** — Jun 2024 – Sep 2024
  Optical Networks and Technologies Lab, NUST — *Islamabad, Pakistan*
  - **Signal ML**: Trained transformer and GAN models to classify optical signals and detect anomalies.

Open-Source Projects

- **TALON** – *Neuromorphic SDK (Python, PyTorch, snnTorch, HLS4ML)* — Docs
  - **SDK**: Built 3 of the SDK's 7 packages (IR, visualizer, simulator), taking spiking neural networks from PyTorch down to FPGA hardware; the visualizer renders **23–30M events/s**.

- **Frames2Py** – *Event-camera streaming library (Python, NumPy, HDF5)* — GitHub | Docs | PyPI
  - **Library**: Turns live event-camera streams into frames that any number of consumers can read without slowing the feed; reads Prophesee EVT 2.0/3.0, AEDAT 4.0 and HDF5.
  - **Performance**: Sustains **20M+ events/s** on pure NumPy across all benchmarks; v1.1.0, Apache-2.0.

Skills

- **Languages:** Python, TypeScript, JavaScript, C++, SQL, Bash
- **AI / LLM:** LLM agents, multi-agent systems, tool calling, RAG, LLM evaluation, inference and serving, vLLM
- **ML:** PyTorch, TensorFlow, transformers, GNNs, SNNs, knowledge graphs, event-based vision
- **Infra / MLOps:** Lambda Labs, GCP, AWS, Azure, Docker, Kubernetes, GitHub Actions, CI/CD, Linux
- **Backend / Data:** FastAPI, Flask, Node.js, REST, SSE, SQLite, MongoDB, MySQL, HDF5

Education

- **B.Eng. in Software Engineering** — Nov 2021 – Jun 2025
  National University of Sciences and Technology (NUST) — *Islamabad, Pakistan*